IRON DEFICIENCY DIET OPTIMIZER

submitted in partial fulfillment of the requirement for the award of the degree

Bachelor of Science

in the faculty of Computer Science with Data Analytics of Bharathiar University, Coimbatore

Submitted by

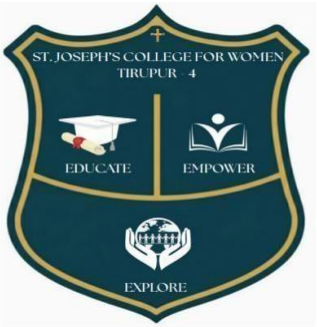
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Department of Computer Science with Data Analytics

ST. JOSEPH'S COLLEGE FOR WOMEN

Affiliated to Bharathiar University

Accredited by NAAC with Grade 'A' (Cycle II)

(Recognized under section 2(f) and 12(B) of the UGC Act, 1956)

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Tirupur – 641 604

OCTOBER - 2026

CERTIFICATE



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This is to certify that the project work entitled **"IRON DEFICIENCY DIET OPTIMIZER"** is submitted to Bharathiar University, Coimbatore in partial fulfillment of requirement for the award of the Degree of Bachelor of Science is a record of the original work done by **H JASMINE FIRTHOUS (2428B0386)** under my Supervision and guidance and that project work has not formed the basis for the award of any Degree/Diploma or similar title to any candidate of any University.

Internal Guide **Head of the Department**

Viva-voce Examination was conducted on _____, in St. Joseph's College for Women, Tirupur – 641604.

Internal Examiner **External Examiner**

Date: **Date:**

DECLARATION

DECLARATION

I, hereby declare that the project work entitled **"IRON DEFICIENCY DIET OPTIMIZER"** is submitted to the Department of Computer Science with Data Analytics, St. Joseph's College for Women, Tirupur affiliated to Bharathiar University, Coimbatore the partial fulfillment of the requirement for the award of Bachelor of Science is an original work done by me during the fifth semester.

Place: **Signature of the Candidate**

Date: **(H JASMINE FIRTHOUS)**

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ACKNOWLEDGEMENT

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SYNOPSIS

SYNOPSIS

The **Iron Deficiency Diet Optimizer** is a web-based application developed using **Python Flask, SQLite, HTML, CSS, JavaScript, and Data Analytics techniques.** The main purpose of the system is to help users monitor their haemoglobin (Hb) levels, receive iron-rich dietary recommendations, and track their diet progress.

The application provides a simple login-based interface where users can **register, log in, and access their dashboard.** Users can enter their health details and haemoglobin test results, view suitable dietary recommendations, and track their daily diet plan. The system also provides charts and reports to analyze haemoglobin records and diet adherence over time.

The system uses **SQLite** to store user information, haemoglobin records, and diet-tracking details. By combining dietary recommendations, progress tracking, and data visualization, the application helps users monitor their dietary habits and manage their progress systematically. It is designed to support health monitoring and does not replace professional medical advice or treatment.

INTRODUCTION

INTRODUCTION

Iron deficiency is a common health concern that can affect haemoglobin levels and overall health. Maintaining a balanced diet with iron-rich foods is important for supporting healthy haemoglobin levels. However, monitoring haemoglobin records, selecting suitable foods, and tracking daily dietary habits can be difficult to manage manually.

The **Iron Deficiency Diet Optimizer** is a web-based application developed using **Python Flask** to help users monitor their haemoglobin (Hb) levels and manage their dietary habits. The system allows users to register, log in, and enter their health details and haemoglobin test results. Based on the user's assessment, the application provides suitable iron-rich dietary recommendations and helps track daily diet progress.

The system also provides data visualization and reports to help users understand their haemoglobin records and diet-tracking progress over time. It uses **SQLite** to store user information, haemoglobin records, and diet-tracking details. The application aims to support users in maintaining healthy dietary habits and monitoring their progress systematically.

ORGANIZATION PROFILE

• ORGANIZATION PROFILE

KICE INFO SYSTEMS is a professional website designing, Customized Software development, Business process Outsourcing – IT/ITES & Internet marketing Company providing full featured web services including B2B Acquisition & B2C Ecommerce solutions and acting as an offshore development center for overseas Development firms. KICE Info systems is an innovative company, based in India that Provides a series of Web-based and software applications that have helped their Customer to create successful business ventures through online initiatives. KICE Info Systems provide all the services that a company needs to get online from web designing to web hosting and manage leading-edge Web sites and e-business applications. Quality and Client Satisfaction are primarily the telling factors of KICE Info systems Success in the domestic market. Ever since its inception, KICE Info systems has Accrued continuous growth in all its business functions and this has been possible only Due to its commitment, quality training methodologies, the services it offers, knowledge Sharing with industry leaders and professional approach. Service Professional Web Design

- SEO concerts, Internet Marketing
- Pay per Click Campaign
- Link Building, Ecommerce Solution
- Web Application Development
- Multimedia Presentations
- Customized Software development
- Business Process Outsourcing- IT/IT

SYSTEM SPECIFICATION

- ## Hardware Configuration

    - **PROCESSOR** : Intel core i3
    - **MEMORY** : 4 GB RAM
    - **STORAGE** : 256 GB
    - **DISPLAY** : 1366X768 resolution
    - **GRAPHICS** : Integrated Graphics
    - **SYSTEM(pc)** **:** Laptop / Desktop

- ## Software Specification

    - **OPERATING SYSTEM (OS)** **:** Windows 11
    - **PYTHON VERSION** **:** Python 3.10
    - **DATASETS** **:** SQLite
    - **FRAMEWORK** **:** Flask
    - **FRONTEND TECHNOLOGIES :** HTML, CSS, JavaScript
    - **PYTHON LIBRARIES**:

        - Flask
        - SQlite3
        - Matplotlib
        - Pandas
        - Numpy

LANGUAGE SPECIFICATION

Python:

The Python language specification is a formal document that defines the syntax, semantics, and grammar of the Python programming language. It serves as the ultimate reference for Python's behavior and provides a detailed explanation of how Python programs should be written and executed. A brief overview of key areas of the Python language specification:

Features:

1. Free and Open Source

Python language is freely available at the official website and you can download it from the given download link below click on the Download Python keyword. Download Python Since it is open-source, this means that source code is also available to the public. So you can download it, use it as well as share it.

2. Easy to code

Python is a high-level programming language. Python is very easy to learn the language as compared to other languages like C, C#, Java script, Java, etc. It is very easy to code in the Python language and anybody can learn Python basics in a few hours or days. It is also a developer-friendly language.

3. Easy to Read

The learning Python is quite simple. As was already established, Python's syntax is really straight forward. The code block is defined by the indentations rather than by semi colons or brackets

4. Object-Oriented Language

One of the key features of Python is Object-Oriented programming. Python supports object-oriented language and concepts of classes, object encapsulation, etc.

5. GUI Programming Support

Graphical User interfaces can be made using a module such as PyQt5, PyQt4, wx Python, or Tk in Python. PyQt5 is the most popular option for creating graphical apps with Python.

6. High-Level Language

Python is a high-level language. When we write programs in Python, we do not need to remember the system architecture, nor do we need to manage the memory.

7. Large Community Support

Python has gained popularity over the years. Our questions are constantly answered by the enormous Stack Over flow community. These websites have already provided answers to many questions about Python, so Python users can consult them as needed.

8. Easy to Debug

Excellent information for mistake tracing. You will be able to quickly identify and correct the majority of your program's issues once you understand how to interpret Python's error traces. Simply by glancing at the code, you can determine what it is designed to perform. Python's language specification encompasses several key features that define its structure and behavior.

PYTHON LIBRARIES:

List of libraries:

- Flask
- SQLite3
- Pandas
- NumPy
- Scikit-learn
- Werkzeug Security

1. Flask:

Flask is a lightweight Python web framework used to develop web applications. In the Iron Deficiency Diet Optimizer, Flask manages user requests, login sessions, health assessment, diet recommendations, diet tracking, and dashboard display.

Key Features:

- **Routing:** Maps URLs to Python functions.
- **Template Rendering:** Displays HTML pages such as Login, Dashboard, Assessment, Diet Plan, Tracker, and Analytics.
- **Request Handling:** Processes user inputs such as login details and Hb values.
- **Session Management:** Maintains the user's login session.
- **Easy Integration:** Works with HTML, CSS, JavaScript, and Python libraries.

2. SQLite3:

SQLite3 is a lightweight database module used to store and manage structured data. In the Iron Deficiency Diet Optimizer, it stores user details, Hb records, and diet-tracking information.

Key Features:

- **Data Storage:** Stores user and health-related records.
- **Data Retrieval:** Retrieves saved information.
- **Data Insertion:** Adds new records to the database.
- **Data Management:** Executes SQL operations on stored records.
- **Easy Integration:** Connects directly with Python applications**.**

Commonly Used Functions

- **sqlite3.connect()** – Connects to an SQLite database.
- **cursor.execute()** – Executes SQL statements.
- **commit()** – Saves database changes.
- **fetchone()** – Retrieves one record.
- **fetchall()** – Retrieves multiple records.

3. Pandas:

Pandas is a Python library used for handling and organizing structured data. In the project, it helps organize the dataset used for training the machine learning model.

Key Features:

- Data Organization: Organizes data into rows and columns.
- Data Processing: Helps prepare data for analysis.
- DataFrame Creation: Creates structured tables from data.
- Feature Management: Organizes input features for machine learning.
- Dataset Analysis: Helps inspect and understand the dataset.

4. NumPy:

NumPy is a Python library used for numerical calculations and array operations. In the project, helps prepare numerical input values for the machine learning model.

Key Features:

- Numerical Operations: Performs calculations on numerical data.
- Array Processing: Stores and processes numerical values.
- Data Preparation: Arranges input values for the model.
- Efficient Computation: Supports fast numerical operations.
- Machine Learning Support: Helps process numerical features.

Commonly Used Functions:

- np.array()
- np.mean()
- np.sum()
- np.max()
- np.min()

5. Scikit-learn:

Scikit-learn is a Python machine learning library used to train and evaluate the classification model in the Iron Deficiency Diet Optimizer. The project uses Logistic Regression with synthetic training data to estimate risk categories; it is not a clinically validated diagnostic too

Key Features:

- **Model Training:** Trains the machine learning model.
- **Data Splitting:** Divides data into training and testing sets.
- **Data Preprocessing:** Scales numerical input features.
- **Classification:** Uses Logistic Regression for classification.
- **Model Evaluation:** Measures the model's performance.

Commonly Used Functions:

- train_test_split()
- Standard Scaler()
- Logistic Regression()
- model.fit()
- model.predict()
- accuracy_score()

Werkzeug Security

Werkzeug Security is used to protect user passwords in the project. Instead of storing passwords as plain text, it helps create password hashes and verify them during login.

Key Features:

- **Password Hashing:** Converts passwords into secure hash values.
- **Password Verification:** Checks whether the entered password matches the stored hash.
- **Login Security:** Supports secure user authentication.
- **Data Protection:** Helps avoid storing plain-text passwords.

Commonly Used Functions:

- generate_password_hash()
- check_password_hash()

Back end: SQLite

Uses: Uses: SQLite3 is a lightweight relational database used for storing and managing structured data. In the Iron Deficiency Diet Optimizer, SQLite3 is used to store and manage user information, haemoglobin (Hb) records, and diet-tracking details. The database is connected to the Flask application using Python's sqlite3 module**.**

- **Data Storage**

  SQLite3 stores user information and health-related records in the database. It maintains details such as user information, Hb values, test dates, test results, and data required for diet tracking.

- **Easy Data Access**

  Flask connects to the SQLite3 database using Python's sqlite3 module. The application can insert and retrieve user information and health records whenever required.

- **Data Management**

  The system stores and manages Hb assessment records and diet-tracking information. This allows users to maintain their records and review previously saved details.

- **Review Analysis Storage**

  SQLite3 maintains stored Hb records and diet-tracking details. This information can be used by the application to display dashboard information, track diet-plan adherence, and present progress through analytics and reports.

Key benefits of SQLite:

- **Readability:**

  SQLite3 stores data in a structured database format. User information, Hb values, test dates, test results, and diet-tracking details can be stored and retrieved easily.

- **Platform Independence:**

  SQLite3 works on different operating systems such as Windows, Linux, and macOS. This makes it suitable for the Flask-based Iron Deficiency Diet Optimizer.

- **Easy Data Storage:**

  SQLite3 works on different operating systems such as Windows, Linux, and macOS.This makes it suitable for the Flask-based Iron Deficiency Diet Optimizer..

- **Easy Data Access:**

  SQLite3 works on different operating systems such as Windows, Linux, and macOS. This makes it suitable for the Flask-based Iron Deficiency Diet Optimizer.

- **Storage Efficiency:**

  SQLite3 is lightweight and does not require a separate database server. The complete database can be maintained in a single database file, making it suitable for this project.

A. SYSTEM STUDY

- **Existing System**

In the existing system, people need to manually monitor their haemoglobin (Hb) levels and search different sources to learn about iron-rich foods and suitable diet plans. Maintaining health records, following a diet plan, and tracking daily food intake can be difficult without a proper system. Users may also find it difficult to compare their current Hb levels with previous records and monitor their diet-plan adherence over time. Additionally, the existing approach may require users to maintain health details and diet records separately, which can make it difficult to organize and review their progress regularly. Finding suitable food recommendations based on individual dietary needs may also take time. Without a centralized system, users may face difficulties in identifying changes in their eating habits, reviewing previous Hb test results, and understanding their overall diet progress. Therefore, a systematic method is needed to maintain health records, organize dietary information, and monitor progress in one place.

- **Drawbacks**
    - Health records are difficult to maintain manually.
    - Finding suitable iron-rich foods can be time-consuming.
    - Tracking daily diet-plan adherence is difficult.
    - Comparing current and previous Hb records requires manual effort.
    - Monitoring diet progress and viewing health-related reports can be challenging.

- ## Proposed System

The proposed system is an Iron Deficiency Diet Optimizer developed using Python Flask. The system allows users to register, log in, enter their health details, and maintain haemoglobin (Hb) records. Based on the information entered, the application provides diet-related guidance and helps users track their diet plans.

The system uses a **Machine Learning model** built with Scikit-learn to estimate risk categories based on the model's input features. The application uses an **SQLite3 database** to store user information, Hb records, and diet-tracking details.

The system provides a dashboard where users can view their health records, follow their diet plans, track adherence, and access analytics and reports. This helps users organize their records and monitor their progress over time. The application supports health monitoring and does not replace professional medical advice or diagnosis.

- ## Features:
  - Maintains user information and haemoglobin records.
  - Provides diet-related guidance and iron-rich food information.
  - Supports daily diet-plan tracking.
  - Displays health records and progress through analytics.
  - Generates reports using stored health and diet-tracking information.

B. SYSTEM DESIGN AND DEVELOPMENT

• File Design

The file design of the Iron Deficiency Diet Optimizer is developed to organize, store, and manage user information and health-related records efficiently. The system maintains details such as user information, haemoglobin (Hb) values, test dates, test results, and diet-tracking information in a well-structured format. This organized file structure helps maintain data consistency, simplifies data processing, and provides quick access to stored health and diet-related information.

The file structure acts as an important component of the system by supporting data processing, storage, and presentation. During execution, the system processes user information using **Python Flask** and stores the required records in an **SQLite3 database**. The application uses separate files for application control, database operations, health assessment, diet planning, tracking, analytics, and report generation. The **app.py** file controls the main application, while the **db.py** file supports database operations. The HTML files such as **login.html, register.html, dashboard.html, assessment.html, diet_plan.html, tracker.html, analytics.html, and report.html** provide the user interface for interacting with the system. The **requirements.txt** file contains the required Python packages for running the application.

- # Input Design

Input Design is one of the important phases in the development of the Iron Deficiency Diet Optimizer. It defines how information is entered into the system by the user. A well-designed input process ensures that the required data is entered correctly and makes the application simple and easy to use. In the Iron Deficiency Diet Optimizer, the input design allows users to enter their personal details, haemoglobin (Hb) values, and diet-tracking information through web forms. The main objectives of input design are:

➤ To reduce errors during data entry.

➤ To provide accurate information for health assessment and diet planning.

➤ To improve the overall efficiency of the system.

The input data is collected through web forms and processed using the **Python Flask** framework. The user details are used for registration and login, while the health information is used for assessment and diet-related guidance. The diet-tracking inputs help users record their progress and monitor their adherence to the suggested diet plan.

User Input:

The project prompts the user to enter the following details:

- **User Details** – Used for registration, login, and maintaining the user session.
- **Health Information** – Used to enter relevant personal and health details for assessment.
- **Haemoglobin (Hb) Value** – Used to record the user's Hb level for monitoring.
- **Test Date and Result** – Used to maintain and review haemoglobin test records.
- **Diet-Tracking Input** – Used to record diet-plan progress and adherence.
- **Login Input** – Used to access the system dashboard.

- # Output Design

Output Design refers to the information and results generated by the Iron Deficiency Diet Optimizer for the end users. The output should be clear, accurate, and easy to understand. In this system, the health-related information and diet-tracking results are displayed through the dashboard in a simple and user-friendly format. The system presents haemoglobin (Hb) records, diet-related guidance, tracking details, and analytics to help users monitor their progress effectively. The output generated by the system helps users review their stored health information and follow their suggested diet plan. The dashboard displays the information in a structured format, making it easier to understand health records and diet adherence.

- ## Output
  - Displays user information and haemoglobin (Hb) records.
  - Shows saved test dates, Hb values, and test results.
  - Provides diet-related guidance and iron-rich food information.
  - Displays diet-plan tracking details to monitor adherence.
  - **Charts** present health records and diet-tracking analytics for easier understanding.
  - **Reports** provide a summary of stored health information and diet-tracking progress.

• Database Design

The most important consideration in designing the database is how information will be used.The main objectives of designing a database are:

- **Data Integration:**

Data Integration is one of the main objectives of the database design. In the Online Product Review Analyzer, product review information collected from different shopping platforms is organized in a structured format. The system maintains details such as username, product name, platform, rating, review, and sentiment in one database.

- **Data Integrity:**

Data Integrity ensures that the information stored in the system is accurate, consistent, and reliable. The system stores review details in a structured format and maintains sentiment results as Positive, Negative, or Neutral. This helps provide reliable review analysis and improves the accuracy of the application.

- **Data Independence:**

Data Independence allows changes to review information or shopping platforms without affecting the overall application structure. New products, reviews, or platforms can be added to the database without changing the basic database design. This makes the application flexible, easy to maintain, and suitable for future enhancements. The data structure of the system is designed to organize product review, rating, sentiment, platform, and visitor information efficiently.

- **System Development**

The Iron Deficiency Diet Optimizer involves data collection, preprocessing, data cleaning, and data analysis to organize and manage health-related information. Data collection involves gathering essential information such as user details, haemoglobin (Hb) test results, test dates, dietary preferences, and daily food intake. This information helps the system maintain health records and support personalized dietary recommendations. Data preprocessing involves organizing the collected information into a suitable format for further processing and use within the application. Data cleaning focuses on identifying invalid, incomplete, or inconsistent inputs to maintain reliable health records and improve the quality of the available data. These steps help prepare the information for effective use in the system. Data analysis is performed to review stored haemoglobin records, monitor daily diet adherence, and understand users' dietary progress over time. The system can display relevant information through charts, summaries, and reports, allowing users to compare previous Hb records and review their diet-tracking progress. Based on the available user details and haemoglobin information, the application also supports suitable iron-rich food recommendations. SQLite is used to store user information, haemoglobin records, and diet-tracking details for future reference. By combining these processes, the Iron Deficiency Diet Optimizer helps users organize their health information, monitor dietary habits, and review their progress systematically. The application is intended to support dietary monitoring and does not re place professional medical advice or treatment.

- **Description of Modules**

The Iron Deficiency Diet Optimizer consists of several modules that work together to provide a health-record and diet-tracking system. The Login and Registration Module allows users to create an account and access the application. The Dashboard Module displays health records and progress information. The Assessment Module processes the user's entered health details. The Diet Plan Module provides diet-related guidance and iron-rich food information. The Diet Tracker Module helps users record their diet-plan adherence. The Analytics Module presents stored information for progress monitoring, while the Report Module provides a summary of health records and diet-tracking information.

1. Login Module

The Login Module is the entry point of the Iron Deficiency Diet Optimizer. It allows registered users to access the application by entering their login credentials. After successful login, the user can access the dashboard and use the available features of the system.

2. Dashboard Module

The Dashboard Module displays the user's health-related information in a simple and organized manner. It provides access to haemoglobin (Hb) records, diet-plan details, tracking information, and analytics. It helps users review their saved information and monitor their progress.

3. Assessment Module

The Assessment Module allows users to enter their personal and health-related details, including haemoglobin (Hb) values. The entered information is processed by the application to provide assessment results and estimated risk categories based on the available model inputs.

4. Diet Plan Module

The Diet Plan Module provides diet-related guidance and information about iron-rich foods. It helps users understand food choices that may support iron intake and follow a suitable diet plan alongside professional medical advice.

5. Diet Tracker Module

The Diet Tracker Module allows users to record their diet-plan progress and monitor their adherence to the suggested plan. The stored tracking information helps users review their food habits and maintain consistency in following the plan.

6. Analytics Module

The Analytics Module organizes stored health records and diet-tracking information for progress monitoring. It presents relevant information in an understandable format, helping users review their records and compare available data over time.

7. Report Module

The Report Module provides a summary of the user's stored health information and diet-tracking details. It helps users review their records and understand their progress in following the diet plan.

8. Machine Learning Assessment Module

The The Machine Learning Assessment Module uses the Scikit-learn library to process input features and estimate risk categories. The model supports the assessment process based on its available training data and is not intended to replace professional medical diagnosis.

9. Database Module

The Database Module uses SQLite3 to store and manage user information, haemoglobin records, and diet-tracking details. It allows the application to insert, retrieve, and manage stored information whenever required.

10. Food Search Module

The Food Search Module allows users to search for food-related information available in the application. It helps users find relevant food details and supports their understanding of iron-rich food choices.

11. Logout Module

The Logout Module allows users to safely exit the application. When the user selects the Logout option, the active session is cleared and the user is redirected from the application.

C. SOFTWARE TESTING AND IMPLEMENTATION

Software Testing is the process of verifying that the Iron Deficiency Diet Optimizer functions correctly and produces the expected results. Testing is performed to identify errors, validate user inputs, verify health assessment and diet-tracking functions, and ensure that the application pages and database operations work correctly. Testing improves the reliability, performance, and usability of the system.

Software Implementation is the process of developing and executing the system using **Python, Flask, HTML, CSS, JavaScript, SQLite3, NumPy, Pandas, and Scikit-learn**. The system processes user registration and login requests, records haemoglobin (Hb) values, provides diet-related guidance, stores health and diet-tracking information in the SQLite3 database, and displays the relevant information through the dashboard.

There are four levels of software testing and implementation was done for this project:

- **Unit Testing:** Tests individual modules such as Login, Assessment, Diet Plan, Diet Tracker, Analytics, and Database separately.
- **Integration Testing:** Checks whether different modules work together correctly and exchange data properly.
- **System Testing:** Tests the complete application to verify that all major functions, including user login, health-record management, diet tracking, and report generation, work correctly.
- **User Acceptance Testing (UAT):** Ensures that the system meets user requirements and provides understandable health-record information and diet-tracking results.

1. Unit Testing:

Unit Testing is the first level of software testing in which each module of the Iron Deficiency Diet Optimizer is tested individually. The main purpose of unit testing is to verify whether each module performs its intended function correctly and produces the expected output. In this project, individual modules such as the Login Module, Assessment Module, Diet Plan Module, Diet Tracker Module, Analytics Module, Database Module, Food Search Module, and Logout Module are checked separately. The Login Module is checked to verify whether users can log in with valid credentials and access the dashboard. The Assessment Module is checked to ensure that the entered health information is processed correctly.

2. Integration Testing:

Integration Testing is performed to ensure that all modules work together correctly without errors. In this project, the interaction between the Login, Assessment, Diet Plan, Diet Tracker, Database, Analytics, and Report modules is checked. For example, after entering haemoglobin (Hb) details, the system should process the information, store the required records in the database, and display the relevant information on the dashboard. Integration testing helps verify proper data flow between different modules of the application.

3. System Testing:

System Testing is performed on the complete Iron Deficiency Diet Optimizer to verify that the entire application functions as a single system. The purpose of system testing is to check the functionality, reliability, usability, and performance of the application. In this project, the major functions, including user registration, login, health assessment, Hb record management, diet guidance, food search, diet tracking, database storage, dashboard display, analytics, report generation, and logout, are checked. The system is evaluated to ensure that users can log in, enter health information, view diet-related guidance, maintain their records, track their diet plans, and access relevant information through the dashboard. System testing helps identify errors and verify the overall functioning of the application.

4. User Acceptance Testing (UAT):

User Acceptance Testing (UAT) is the final stage of software testing used to check whether the Iron Deficiency Diet Optimizer meets user requirements. The system is evaluated for successful registration and login, health information entry, diet guidance, food search, diet tracking, and proper display of analytics and reports. Users check whether the application is easy to use and provides understandable health records and diet-related information. The dashboard, assessment results, diet-plan details, and database functions are also considered during evaluation. UAT helps determine whether the application is suitable for its intended purpose. The system supports health monitoring and diet tracking and does not replace professional medical advice or diagnosis.

CONCLUSION

CONCLUSION

The Iron Deficiency Diet Optimizer provides a simple way to monitor haemoglobin (Hb) records and manage iron-rich dietary habits. The system uses Python Flask, SQLite, HTML, CSS, and JavaScript to provide health assessment, diet guidance, diet tracking, and analytics features. The dashboard presents users' health records and diet-related information clearly, helping them monitor their progress and maintain their diet plan. Overall, the system reduces the effort required for manual health record tracking and provides a useful solution for monitoring dietary habits related to iron deficiency. However, it supports health monitoring and does not replace professional medical advice or diagnosis.

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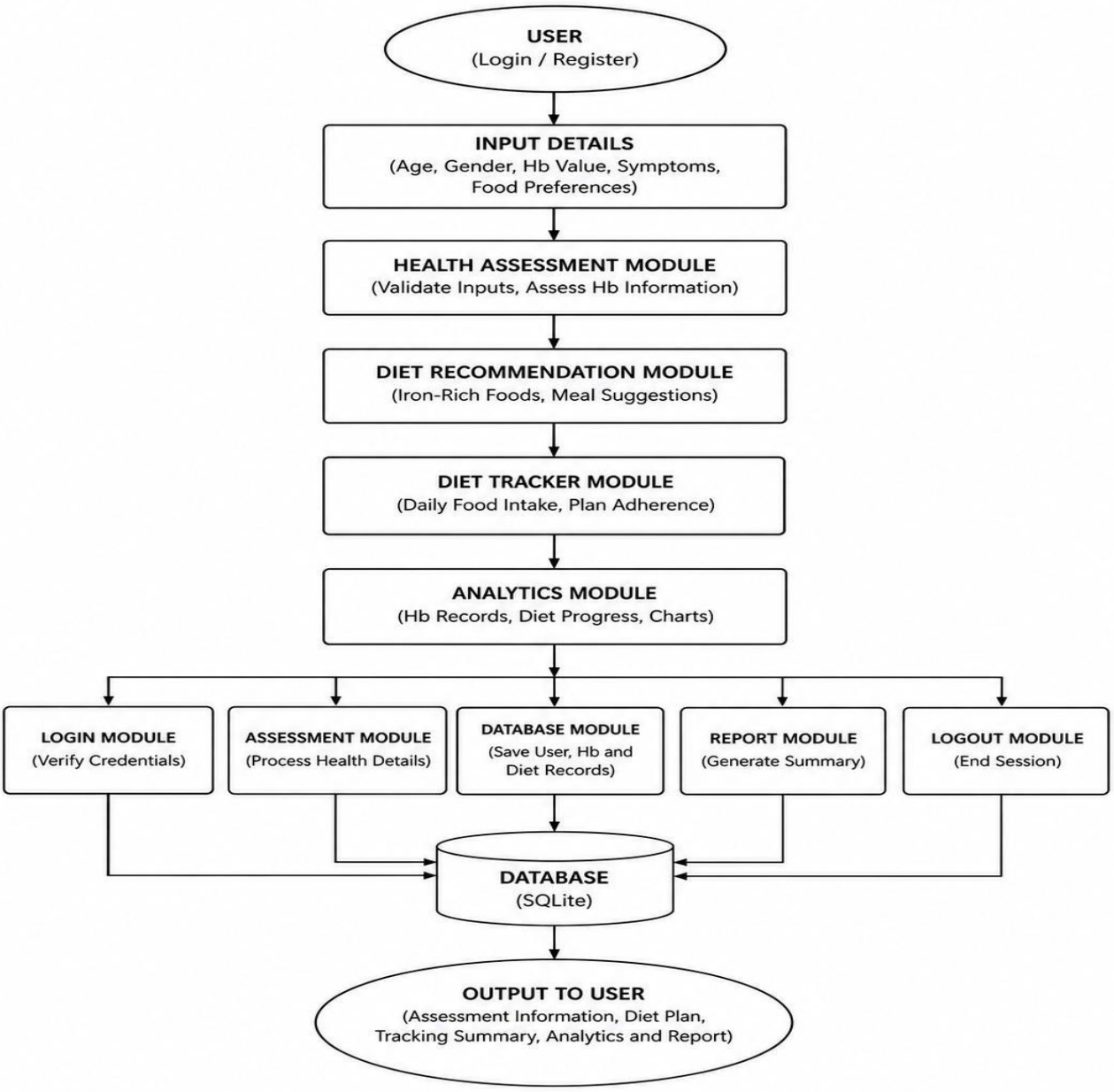
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    - https://www.sqlite.org/docs.html
    - https://www.w3schools.com/html/
    - https://www.w3schools.com/css/
    - https://pandas.pydata.org/docs/
    - https://numpy.org/doc/
    - https://scikit-learn.org/stable/
    - https://ods.od.nih.gov/
    - https://www.who.int/

APPENDIXES

D. Data Flow Diagram

APPENDIXES

D. Data Flow Diagram



APPENDIXES

E. Table Structure

D. Table Structure

1. User Details Table

Field Name	Data Type	Description
id	INTEGER	Unique ID for each user
name	TEXT	Full name of the user
username	TEXT	Unique username used for login
password	TEXT	Password stored in hashed form
created_at	TEXT	Account creation date and time
age	INTEGER	Age of the user
gender	TEXT	Male / Female
height	REAL	Height in cm
weight	REAL	Weight in kg
diet_preference	TEXT	Vegetarian / Non-Vegetarian
activity_level	TEXT	Low / Moderate / High
medical_conditions	TEXT	Medical conditions of the user

2. Health Records Table

Field Name	Data Type	Description
id	INTEGER	Unique ID for each record
user_id	INTEGER	User ID (links to User Details table)
test_date	TEXT	Date of Hb test / assessment
hb_value	REAL	Haemoglobin value in g/dL
result	TEXT	Low / Moderate / High risk
source	TEXT	Manual / ML model / symptoms
date	TEXT	Date of the diet log
breakfast	INTEGER	Breakfast taken (1) or not (0)
lunch	INTEGER	Lunch taken (1) or not (0)
dinner	INTEGER	Dinner taken (1) or not (0)
snacks	INTEGER	Snacks taken (1) or not (0)
water_intake	INTEGER	Glasses of water consumed
score	REAL	Daily diet score (%)

APPENDIXES

F. Sample Coding

E. Sample Coding

1. Database Connection

```
def get_db():
    conn = sqlite3.connect(DB_PATH)
    conn.row_factory = sqlite3.Row
    return conn
```

2. Daily Diet Score Calculation

```
        meals_eaten = breakfast + lunch + dinner + snacks


score = round(
    (meals_eaten / 4 * 70)
    + (min(water, 8) / 8 * 30), 1
)
```

3. Goal Progress Calculation

```
        TARGET_HB = 12.0
def goal_progress(first_hb, current_hb):
    if first_hb is None or current_hb is None:
        return 0

    if first_hb >= TARGET_HB:
        return 100 if current_hb >= TARGET_HB else 0
    progress = (
        (current_hb - first_hb)
        / (TARGET_HB - first_hb) * 100
    )
return max(0, min(100, round(progress)))
```

4. Iron-Rich Food Recommendation

```
 for meal in ["Breakfast", "Lunch", "Dinner",
"Snacks"]:
    if food_preference == "Non-Vegetarian":
        items = [
            dict(f) for f in FOOD_DATA
            if f["meal"] == meal
        ]
    else:
        items = [
            dict(f) for f in FOOD_DATA
    if f["meal"] == meal
    and f["food"] not in [
        "Boiled Egg",
        "Chicken Curry",
        "Fish Curry",
```

```
            "Mutton Curry",
            "Grilled Fish"
        ]
if risk in ("High", "Moderate"):
    items.sort(
        key=lambda f: f["iron_mg"],
        reverse=True
```

5. Haemoglobin Risk Classification

```
def classify_hb(hb_value, gender="Female"):
    low_cut = 12.0 if gender == "Female" else 13.0
    high_cut = 8.0

    if hb_value >= low_cut:
        return "Low"
    elif hb_value >= high_cut:
        return "Moderate"
```

```
else:
    return "High"
```

6. BMI Calculator

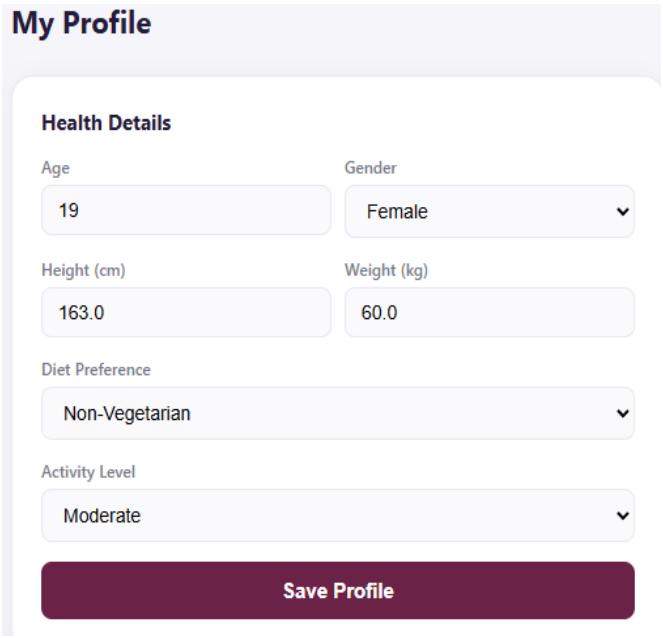
```
const hm = h / 100;
const bmi = w / (hm * hm);

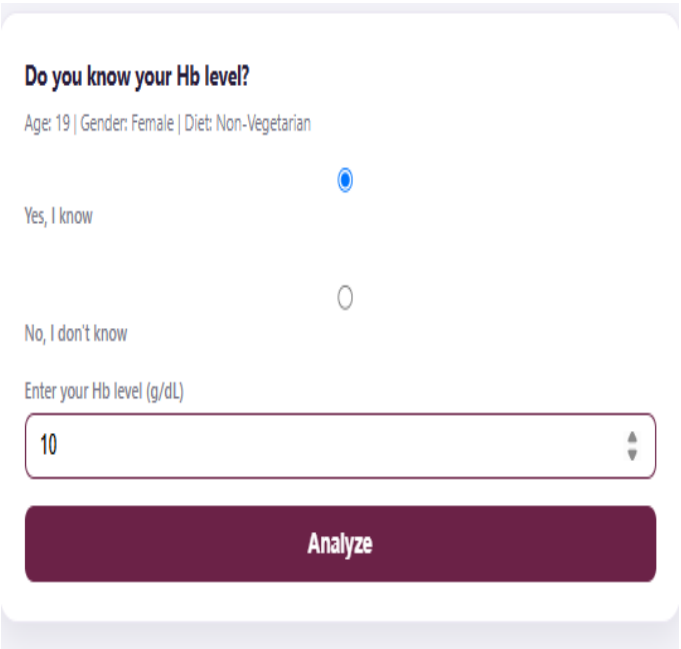
if (bmi < 18.5) cat = "Underweight";
else if (bmi < 25) cat = "Normal";
else if (bmi < 30) cat = "Overweight";
else cat = "Obese";
```

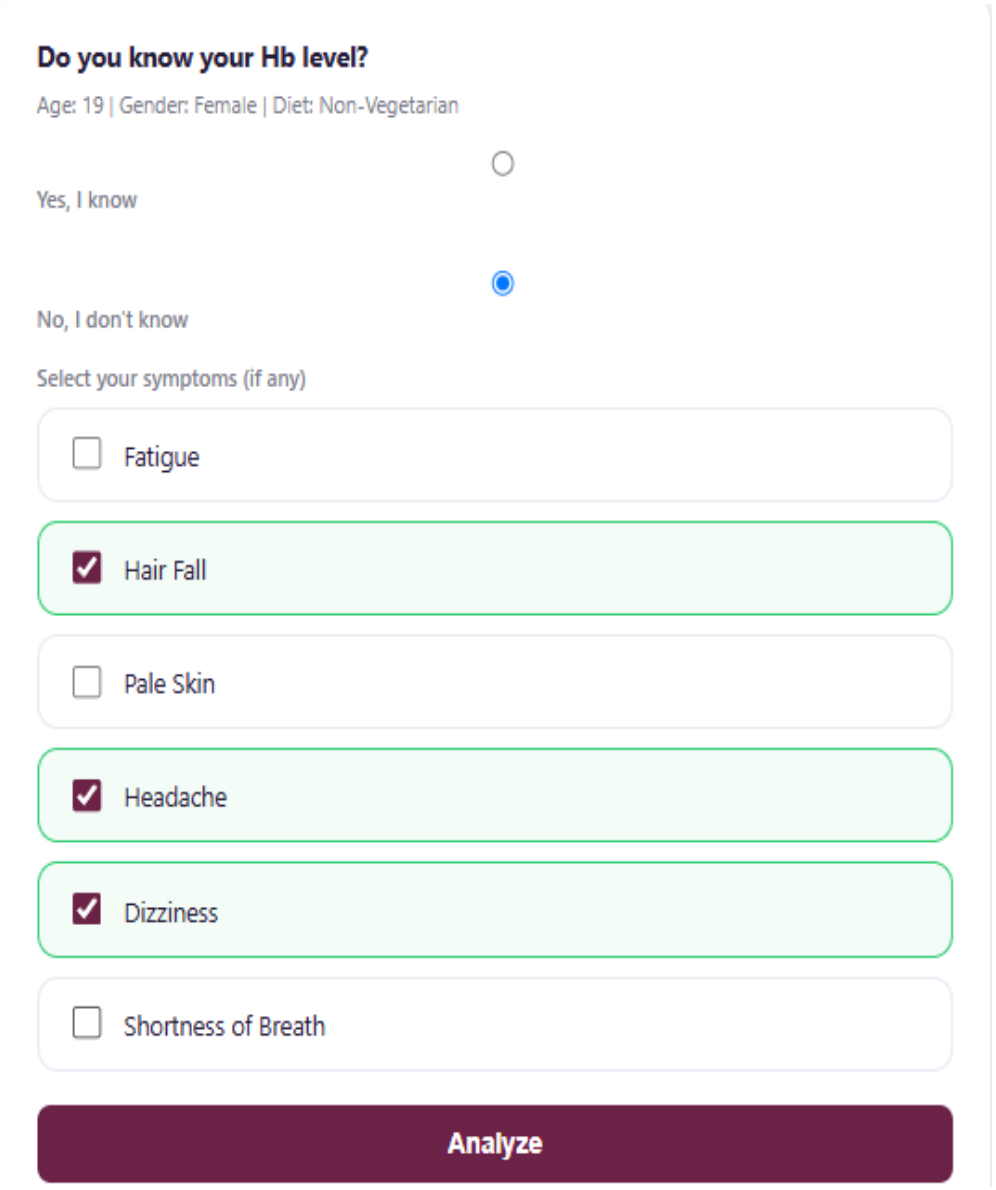
APPENDIXES

G. Sample Input

F. Sample Input

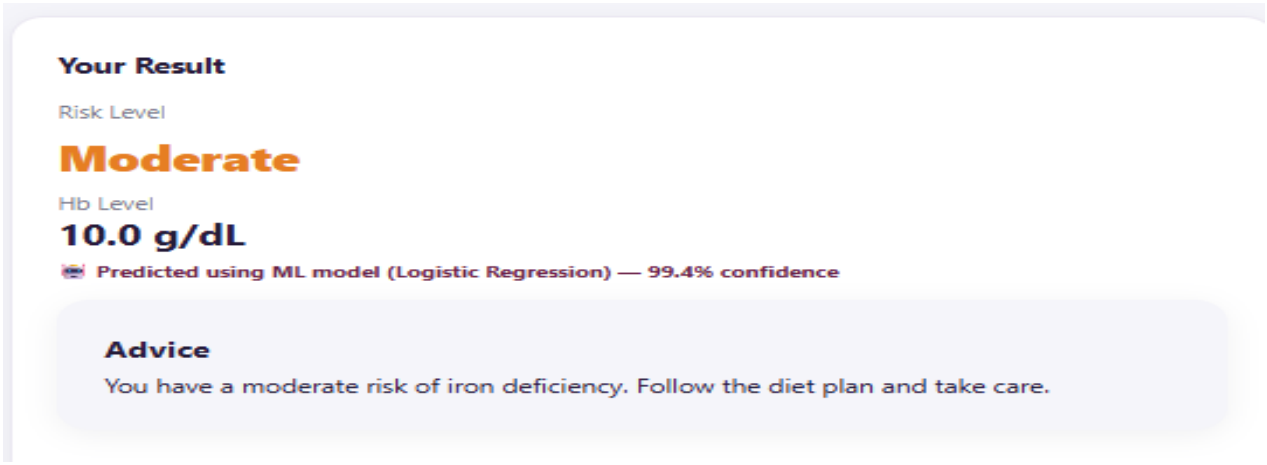


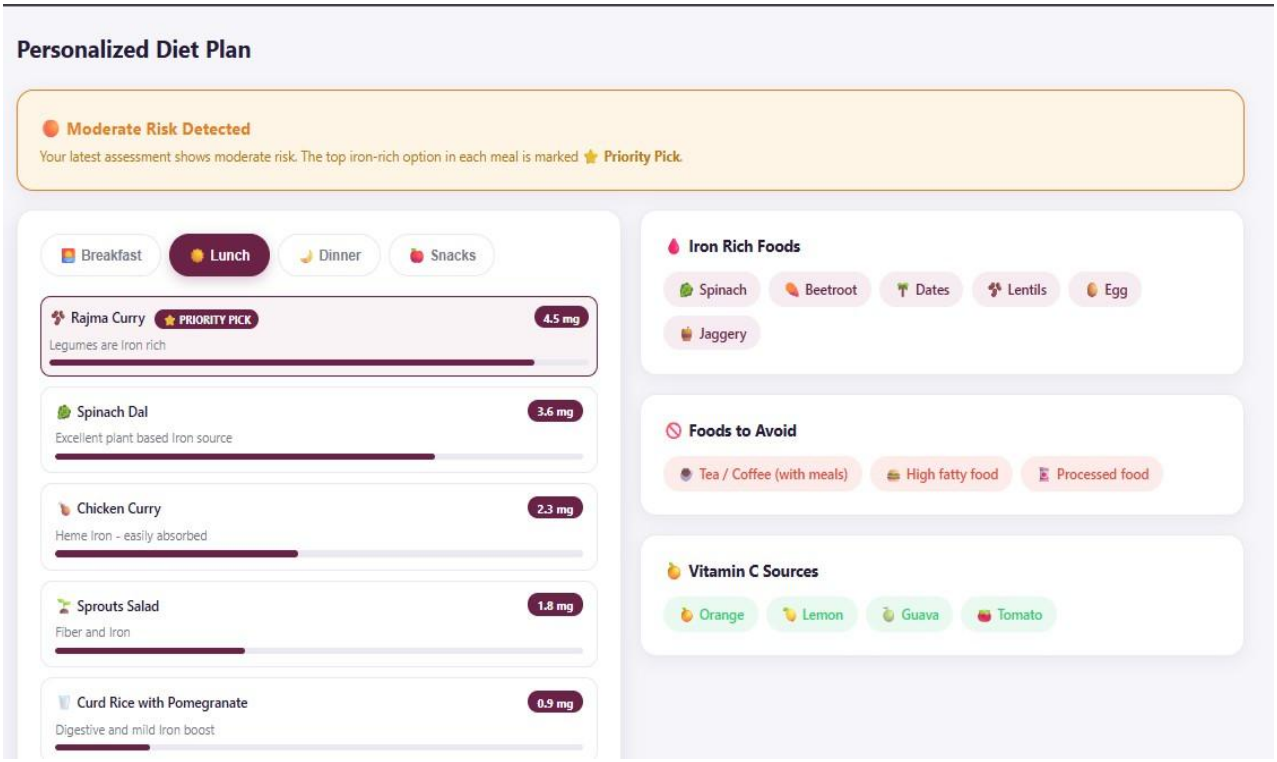


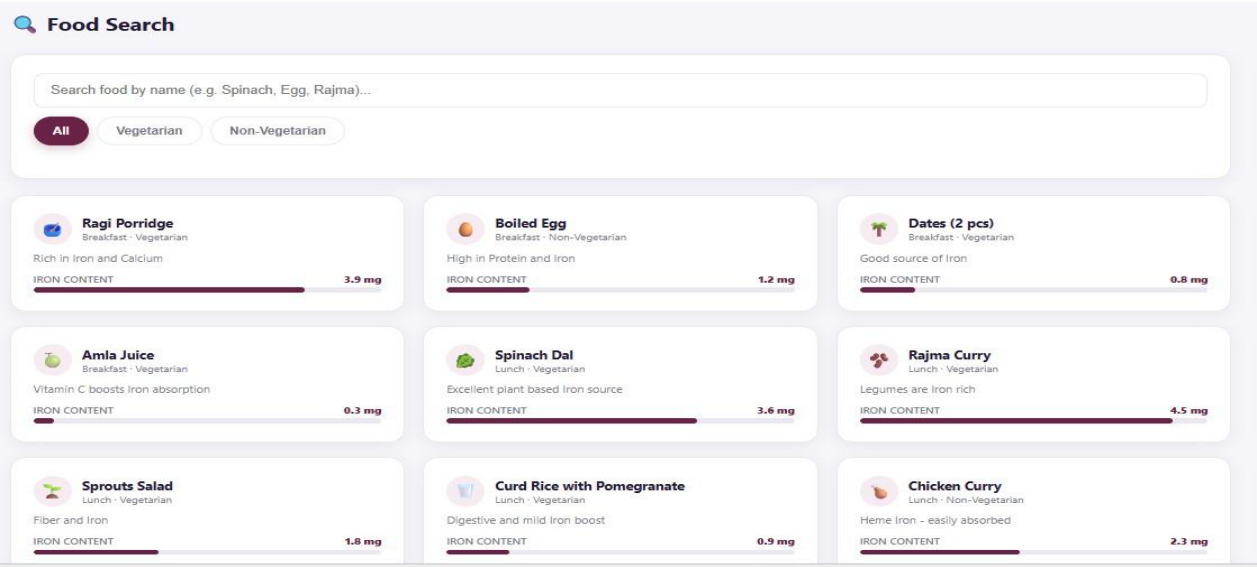

Do you know your Hb level?
Age: 19 | Gender: Female | Diet: Non-Vegetarian
Yes, I know
No, I don't know
Select your symptoms (if any)
Fatigue
Hair Fall
Pale Skin
Headache
Dizziness
Shortness of Breath
Analyze

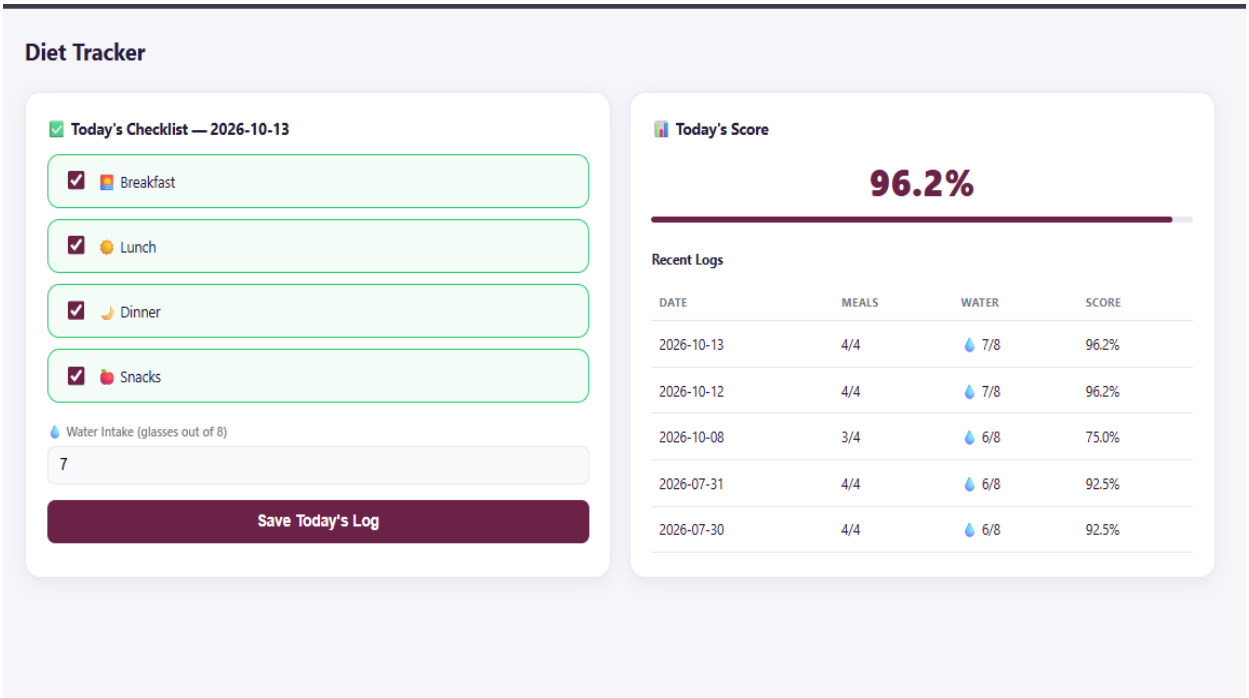
H. Sample Output

G. Sample Output

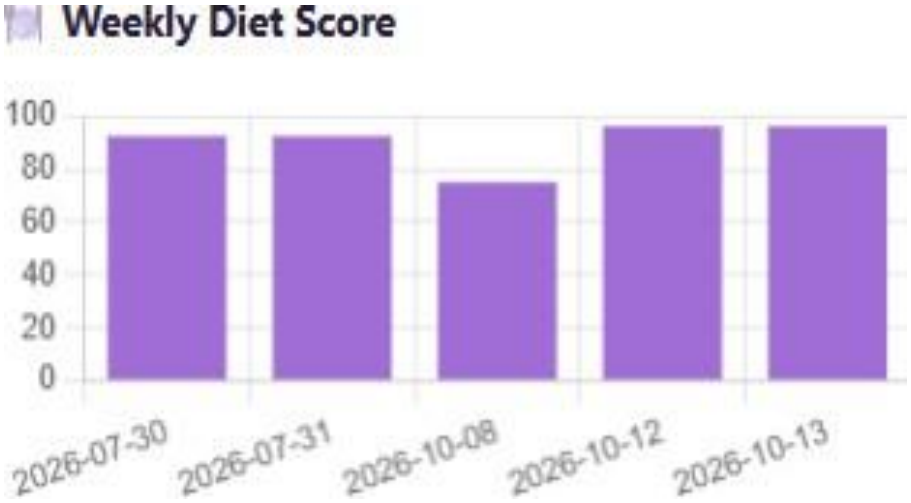


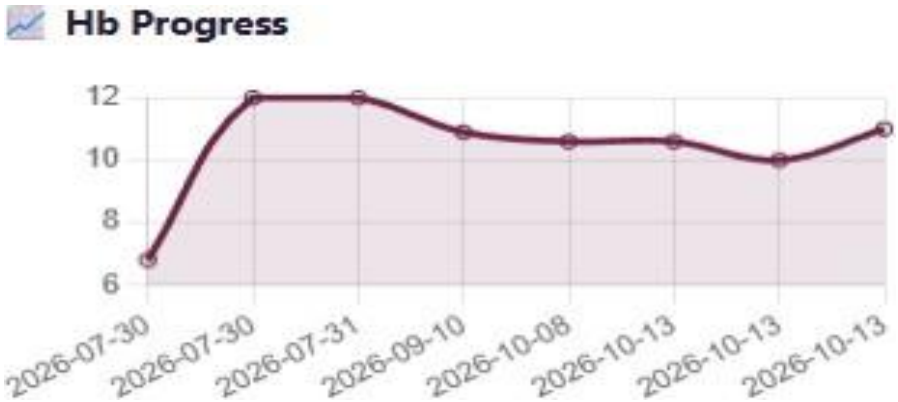


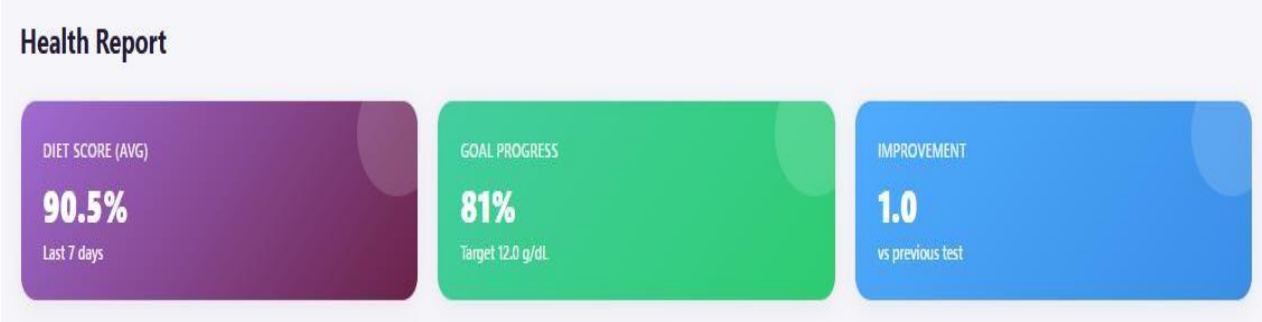

Food Search
Search food by name (e.g. Spinach, Egg, Rajma)...
All
Vegetarian
Non-Vegetarian
Ragi Porridge
Breakfast · Vegetarian
Rich in Iron and Calcium
IRON CONTENT
3.9 mg
Boiled Egg
Breakfast · Non-Vegetarian
High in Protein and Iron
IRON CONTENT
1.2 mg
Dates (2 pcs)
Breakfast · Vegetarian
Good source of Iron
IRON CONTENT
0.8 mg
Amla Juice
Breakfast · Vegetarian
Vitamin C boosts Iron absorption
IRON CONTENT
0.3 mg
Spinach Dal
Lunch · Vegetarian
Excellent plant based Iron source
IRON CONTENT
3.6 mg
Rajma Curry
Lunch · Vegetarian
Legumes are Iron rich
IRON CONTENT
4.5 mg
Sprouts Salad
Lunch · Vegetarian
Fiber and Iron
IRON CONTENT
1.8 mg
Curd Rice with Pomegranate
Lunch · Vegetarian
Digestive and mild Iron boost
IRON CONTENT
0.9 mg
Chicken Curry
Lunch · Non-Vegetarian
Heme Iron - easily absorbed
IRON CONTENT
2.3 mg


Diet Tracker
Today's Checklist — 2026-10-13
Breakfast
Lunch
Dinner
Snacks
Water Intake (glasses out of 8)
7
Save Today's Log
Today's Score
96.2%
Recent Logs
DATE
MEALS
WATER
SCORE
2026-10-13
4/4
7/8
96.2%
2026-10-12
4/4
7/8
96.2%
2026-10-08
3/4
6/8
75.0%
2026-07-31
4/4
6/8
92.5%
2026-07-30
4/4
6/8
92.5%

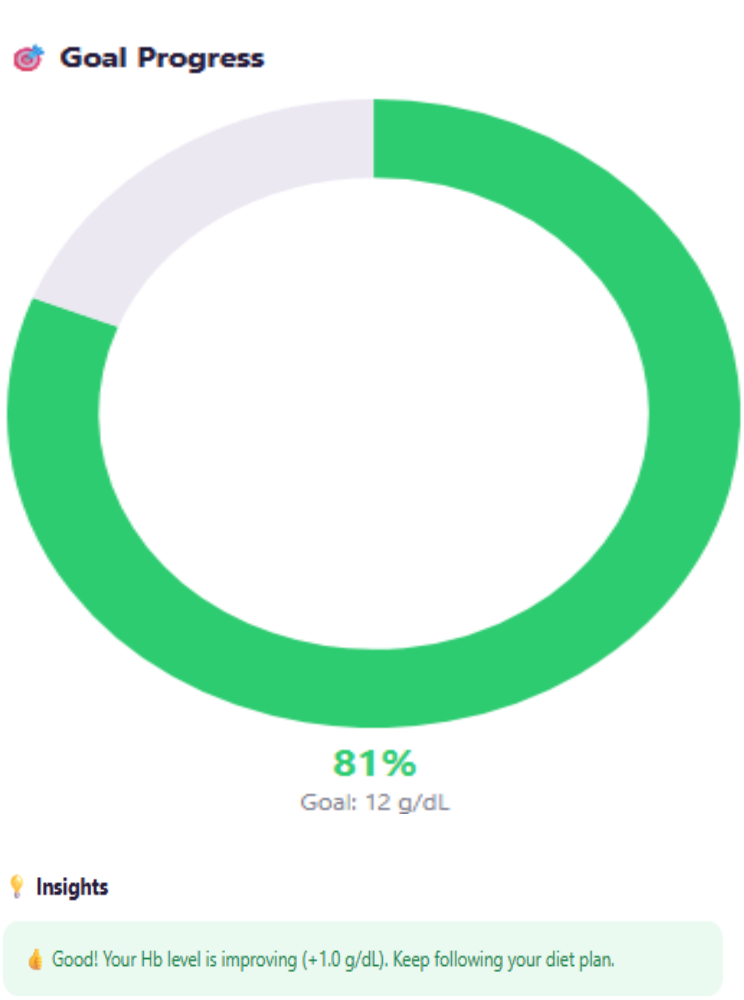
DASHBOARD: BAR CHART, LINE CHART






Health Report
DIET SCORE (AVG)
90.5%
Last 7 days
GOAL PROGRESS
81%
Target 12.0 g/dL
IMPROVEMENT
1.0
vs previous test

Donut chart


Goal Progress
81%
Goal: 12 g/dL
Insights
Good! Your Hb level is improving (+1.0 g/dL). Keep following your diet plan.